Addressing the Ethical Implications of AI Models Developed: A Case Study of Master's Degree Dissertations in Data Science for Industry and Society

Keywords: Ethical AI, Data Science, Responsible AI, Bias, Transparency, Accountability, Fairness, Explainable AI, Safety, Data Privacy, Machine Learning, Artificial Intelligence

Abstract: Artificial intelligence (AI) is an emerging field with a wide range of applications in many fields. It is one of the most widely used technologies in the field of computer science, especially in the areas of health, education, and medicine. The increase in the development and use of AI models has generated many ethical and societal concerns. In this paper, we will examine the ethical element in several dissertations presented in July and September 2024 by students enroled in the Data Science for Industry and Society Master's Degree Programme. The focus of this study is to assess the level of awareness of ethical principles related to developing consumer-oriented AI applications. We analyse the levels of ethical AI literacy among these case studies, the ethical concerns addressed by most students, the ethical principles that are mostly neglected, and possible implications for the society. The findings reveal that data bias is the most addressed concern, while accountability is the most neglected ethical principle. Some recommendations for possible improvements, based on the existing literature, are suggested in the Discussion section.

1 INTRODUCTION

The past decade has known an exponential growth of Artificial Intelligence (AI) based applications. Many of these, for example public Large Language Models (LLM), introduce disruptive technologies that challenge and modify for good many aspects of life. The activities involved include teaching and learning, producing written content such as essays, reports, or news, and constitute a huge knowledge and decision base for many users. Despite many disclaimers from the manufacturers of these tools explaining that erroneous content might be generated, many users are trustingly using them in professional and learning contexts. Despite being scrutinised, the use of these models in education to produce writing that is creative, cohesive, and pertinent is deeply explored, emphasising their role in improving the quality and efficiency of student work (Kenwright, 2024).

More often than not, for many such applications, the level of accountability demonstrated by the developer is often neglected, with the result that the user has the primary responsibility to use the AI (Vakkuri, 2022). Caution and reluctance are often advised because the ethical and safety implications are not made known to the user, who is at high risk. Rather than spending too much controversial time on whether we should use and trust these systems or not in educational, medical, or any other important area of our lives, we might focus more productively on developing ethical models. Promoting transparency, agreeing on safety and inclusive guidelines to be enforced could finally make this technology serve actual societal needs and improve life quality (Remian, 2019).

The quality of AI is determined by the availability and quality of data, which supports the claim that data is the primary source of AI development, and it is where the majority of problems originate. It is critically important to ensure that AI systems are trained on a variety of representative datasets and that bias is rigorously tested in them (Konidena et al., 2024). However, sometimes it is this data that mirrors for us the hidden biases we have been socially propagating unaware, that actually draws attention to the issue. Going even further, it has been demonstrated that AI attacks that utilise gradient-based enhancements are possible (Rosenblatt et al., 2023) resulting in falsely highly accurate systems, with the result of fraudulent AI models being used based on unreal claims.

Given all these issues and problems raised, the ethical component of our intelligent applications is becoming more and more relevant. There has been

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a growth in the last 10 years in research on the relevant aspects of ethical and responsible AI, and their priority. The correct identification of these concerns is the first step in addressing them. For example, some researchers apparently argue that unemployment as a result of the automation power of AI models is one of the main ethical implications (Wiesenthal, 2022). This calls for a shift in mindset and culture, as this particular concern can be handled by gradually retraining individuals for other higher-level and more engaging occupations, while gradually introducing automation technology. There are several ethical concerns, which we are still not ready to fully address, which are even more critical. Moreover, even if ethical concerns are identified and discussed, the majority of research does not offer recommendations on how to deal with these problems when designing and developing AI systems (Huriye, 2023).

There are researchers (Balasubramaniam, 2019) that analyse how engineering requirements can help businesses handle ethical concerns with AI technologies. They examined three Finnish firms' AI ethics guidelines using a multiple-case research methodology. Accountability, justice, privacy, safety, security, transparency, and trust were the main topics of the recommendations. Two of the companies suggested creating multidisciplinary development teams to address divergent ethical perspectives. The findings imply that businesses might prioritise quality requirements by using AI ethical norms.

The literature now in publication has recognised the following major categories of ethical issues: privacy, autonomy, anonymity, transparency, security, safety, justice, and dignity (Tatineni, 2019). These should normally come on top of software engineering quality requirements of usability, performance, reliability, security, safety, maintainability, accuracy, interoperability, and reusability (Balasubramaniam, 2019).

Real-world case studies offer valuable insight into the successful integration of ethical considerations in AI and Data Science (Tatineni, 2019). Examples include proactive bias mitigation, transparent decision making, and community participation. These strategies ensure fair outcomes for diverse user groups, foster trust, and align ethical considerations with societal values, thereby fostering a more inclusive and ethical AI landscape.

The ethical considerations in AI and Data Science development are crucial for responsible use. Strategies include conducting ethical impact assessments, developing agile ethical frameworks, and fostering collaboration between researchers, ethicists, policymakers, and industry experts. By understanding the implications of advanced AI, anticipating future challenges, and implementing proactive measures, we can contribute to responsible and beneficial technology development.

The aim of this paper is to analyse the existing level of knowledge and awareness of ethical AI implications among master's degree students from a specialised Data Science domain. Several thesis are being assessed on 4 selected ethical criteria: (1) fairness and bias, (2) safety and security, (3) accountability and liability, and (4) transparency and explainability. The issues concerning these aspects and how they are addressed by students or could be addressed in the future is systematised in the following research questions (RQ):

- **RQ1: Which ethical principles for developing AI-based applications are most often addressed by students?**
- **RQ2: Which ethical principles for developing AI-based applications are generally neglected by students?**
- **RQ3: What solutions for the implementation of ethical AI principles from the literature could be applied in these case studies?**

2 MATERIALS AND METHODS

2.1 Methodology

In this study, we evaluate the ethical element in several dissertation theses presented in July and September 2024 by students enroled in the Data Science for Industry and Society Master's Degree Programme. The criteria used in the analysis of these case studies are: (1) fairness and bias, (2) safety and security, (3) accountability and liability, and (4) transparency and explainability. All identified concerns are then assessed as to how they are addressed in the paper, if at all, by means of: discussion, evaluation tools utilised, ethical design, and other approaches.

From a number of 12 student papers, we selected those that involve either the development of an AI module or the development of an application that uses pre-trained AI models. Papers that do not have as scope a specific application addressed to a large pool or users, but are focused more on research methods, comparisons, or assessment of AI models have been excluded. The focus of this study is to assess the level of awareness and mitigation of ethical issues related to developing consumer-oriented applications. As these have a very high impact on our current society, the question is if the people developing the tech-

nology that shapes our lives are aware of the impact they have through the AI they use and develop.

The methodology is presented in Figure 1. The selected dissertations are being assessed considering the four criteria that are derived from both the related literature, and from ethical AI norms and recommendations, by institutions such as UNESCO (UNESCO, 2024). The criteria will be analysed by identifying several concerns specific to each AI model or application, and evaluating if and how they are identified, addressed, and mitigated in the paper. In the end, we will analyse the level of ethical AI literacy among these students and possible implications, suggesting some recommendations where improvement is possible, based on the literature.

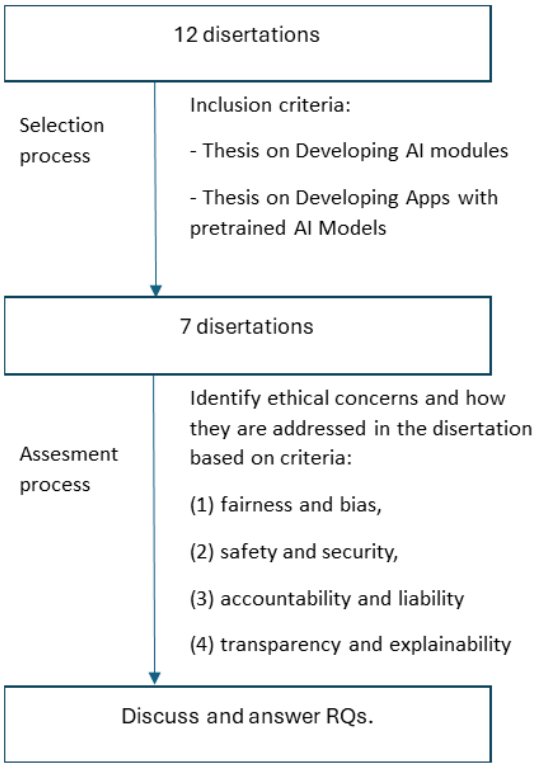


Figure 1: The methodology schema.

2.2 Ethical AI principles

The criteria chosen for this study are based on related literature (Horváth, 2022), (Konidena et al., 2024), (Nguyen et al., 2023) and in line with the recommendations provided by UNESCO's "Ethics of Artificial Intelligence" (UNESCO, 2024). The complete list includes several aspects which are related also to the Effects on Society and the Economy (Sustainability), user Confidentiality and Privacy (which is a mode delicate ethical discussion) and system Autonomy (Human agency and oversight), which we will discuss in relation to the remaining four criteria mentioned up to this point. These are described in detail in Figure 2, in order to have a clear understanding of what we are focusing on in the analysis of the thesis and identification of possible concerns.

2.3 Summaries of the selected papers

2.3.1 CS1: Speed Bump Detection

Case study 1 (CS1): Speed Bump Detection (Complete Title: Robust Speed Bump Detection Based on Data Collected from GPS-Enabled Dashcams) (Muntean, 2024). Using predetermined GPS locations, this dissertation proposes a novel way for Advanced Driver Assistance Systems (ADAS) to gather speed bump data. A labelled dataset is constructed with 2351 photos of plastic speed bumps that are black and yellow, used to train a speed bump detection AI model. The research shows that, while maintaining a lower False Positive Rate, YOLOv8, a cutting-edge object identification model, can achieve performance levels comparable to those of the literature. The authors claim that the new model is among the best models for speed bump detection available, having been pre-trained on the higher version of YOLOv8 and achieving a false positive rate of only 3.4%.

2.3.2 CS2: Dental Issues Identification

Case study 2 (CS2): Dental Issues Identification (Complete Title: Advancements in Dental Care through Deep Learning) (Moldovan, 2024). In order to detect frequent dental problems including cavities, implants, impacted teeth, and fillings, this study examines a number of object identification models, including pre-trained ResNet models and custom models trained with TensorFlow Keras. YOLOv9 and other YOLO (You Only Look Once) models are tested for real-time processing on mobile devices. In a mobile application setting, the study demonstrates the effectiveness of YOLO models for dental X-ray processing, which could expedite patient sessions and provide users insights into their oral health. With an F1 score of 0.8733, recall of 0.8502, and overall precision of 0.8974, the authors conclude that the YOLOv9 model performs better than other bespoke models.

2.3.3 CS3: News Analysis

Case study 3 (CS3): News Analysis (Complete Title: Beyond the Headlines: Leveraging AI to Streamline

Fairness and bias	To guarantee that everyone can benefit from AI, AI actors should advocate for social justice, fairness, and nondiscrimination while adopting an inclusive stance. Aim to identify in data the social groups less well represented or presenting prejudices, to mitigate these issues and promote fairness.
Safety and security	Take precautions against unwanted consequences (safety risks) and attack weaknesses (security risks). The AI lifecycle should emphasize safeguarding privacy, alongside developing robust data protection structures. Aim to identify and itigate possible misuse and risks.
Accountability and liability	Mechanisms for supervision, impact assessments, audits, and due diligence should be in place. Aim such that AI systems do not eliminate the ultimate responsibility and accountability held by humans. In the realm of AI, programmers, data providers, system administrators, and end users alike could all bear collective responsibility.
Transparency and explainability	AI systems' ethical use hinges on their explainability and transparency. Aim to provide a transparent decision making process of the AI model, as well as being open on data sources utilised, development and robustnesss of the model, system limitations and possible secondary impact to environment, to promote trustworthy AI. It is essential to thoroughly consider human agency and control during both the design and deployment of AI systems.

Figure 2: The Ethical AI criteria used for assessment.

News Analysis) (Miclea, 2024). The project's goal is to create an intelligent news aggregation and analysis system that uses artificial intelligence and natural language processing to automate the task. The system consists of a sentiment analysis module, data processing engine, news retrieval module, and user interface. The author claims that initial findings demonstrate enhanced sentiment analysis instruments and fixed clustering in data classification, but the evaluation criteria are not very clear. The future work of the project aims to improve sentiment analysis to capture subtle emotional tones and refine news grouping to better match the subjects. Incorporating user feedback for customised and improved news aggregation is another future goal of the project.

2.3.4 CS4: Anime recommendation

Case study 4 (CS4): Anime recommendation (Complete Title: Anime recommendation system) (Lăzărescu, 2024). This thesis compares the results of experiments conducted on a small real-world dataset with the complete MyAnimeList database for the purpose of designing an application to generate anime recommendation. To determine which model was the most effective and recommended, three different models were evaluated: NCF,Wide & Deep, and SVD, with the best result for $R^2 = 0.5473$. The authors conclude that their findings offer strong proof that the recommendation models presented are suitable for developing an application that can correctly identify and suggest anime to users according to their tastes.

2.3.5 CS5: Stock trend prediction

Case study 5 (CS5): Stock trend prediction (Complete Title: Predicting Romanian stock movement trends: a complex network approach combined with machine learning) (Holgyes, 2024). This study concentrates on predicting the trend movements of the Bucharest Stock Exchange (BSE). It examines stock price data through the lens of network science principles to uncover insights into how the BSE operates. Next, specific features are derived from the network model, including centrality measures and connectivity properties, in order to train several regression algorithms. The top performance was achieved using the decision tree model, which predicts the following day's price movements (rise, fall, or stagnate) with an accuracy exceeding 70% and a precision surpassing 50%.

2.3.6 CS6: Blood cell classification

Case study 6 (CS6): Blood cell classification (Complete Title: Blood cells classification for Acute Lym-

phoblastic Leukemia detection using federated learning) (Chiorean, 2024). This research investigates the application of federated learning for classifying white blood cells to detect Acute Lymphoblastic Leukemia (ALL). The paper reviews current techniques and their uses, especially Convolutional Neural Networks (CNNs) and the advancements provided by MobileNet and ResNet architectures. The findings indicate that federated learning can combine knowledge from multiple clients. In addition, maintaining stability and careful adjustment of the hyperparameters are essential for optimal outcomes. The model was able to improve performance up to 90% for accuracy and 80% for precision. ResNet demonstrates superior precision and reliability in diagnosing ALL compared to MobileNet and CustomNet, while MobileNet excels in performance with a smaller number of parameters.

2.3.7 CS7: Travel recommendation

Case study 7 (CS7): Travel recommendation (Complete Title: A Travel Recommendation System Based on Weather Data and Traveller Profile Using Machine Learning Algorithms) (Zbârcea, 2024). This dissertation aims to assist users in choosing the ideal vacation spot tailored to their individual preferences and requirements. It introduces a web-based platform that provides customised recommendations, valuable insights, and weather-related functionalities. The system employs a hybrid approach, utilising Long Short-Term Memory models for predicting weather conditions in conjunction with a deep learning model trained on a dataset of virtual users with their personal preferences and past travel experiences.

3 RESULTS AND DISCUSSION

3.1 Analysing ethical AI concerns

A total of **83** ethical AI concerns have been identified from these papers, however, some are similar issues which are present for many of the use cases. One paper of the seven, namely CS1, is distinguished as having addressed most of the safety and security and some of the fairness and bias related issues (9 issues out of 21), while the others fail to address ethical concerns in most aspects, as described in Figures 3 and 4. Also, CS1 has the most ethical concerns addressed in total, as it is the most detailed work and also perhaps because the application domain (automotive) is a popular field worldwide. The percentage of issues identified per case study and criteria are presented in Figures 5 and 6, showing that the majority of concerns are related to fairness and bias, and the fewest are related to accountability and liability. This is perhaps because data are a volatile resource in which many different aspects can generate ethical issues like biases or under-represented populations, that will influence the model in various ways. Although accountability is not less important, it is a more straightforward issue of who is responsible for the technology functionalities and its use.

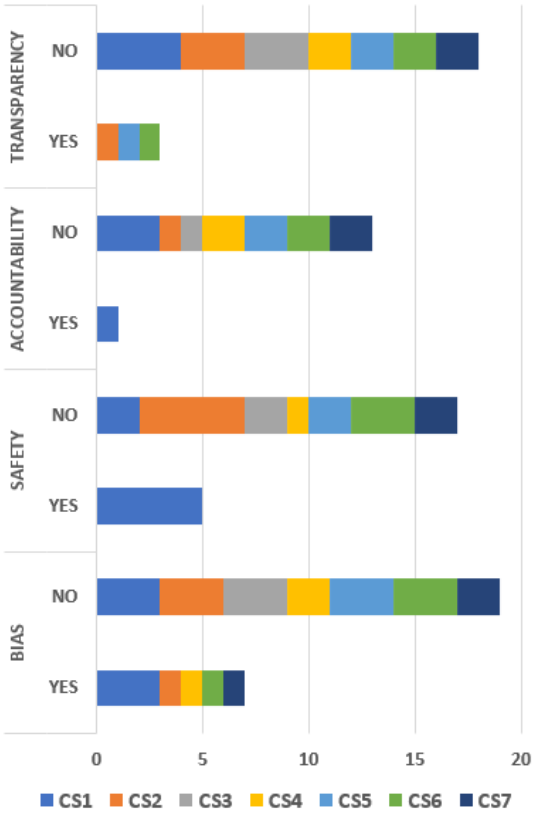


Figure 3: Number of addressed concerns per criteria for each case study.

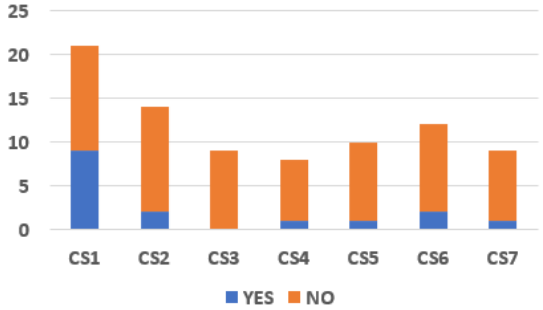


Figure 4: Number of concerns for each case study (yes - addressed, no -not addressed).

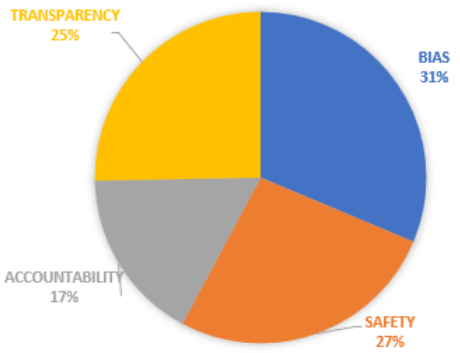


Figure 5: Distribution of concerns per criteria .

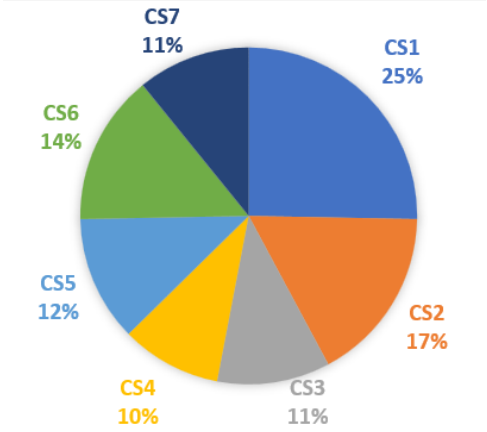


Figure 6: Distribution of concerns per case study.

In Table 1 we present in detail several concerns related to fairness and bias, as they were identified in each of the 7 dissertations, and if they are addressed in any way (either just identified, perhaps discussed, or even mitigated) by the student.

A similar analysis for AI safety and security is performed in Table 2. Table 3 presents the issues related to accountability and liability of AI systems, while Table 4 is concerned with transparency and explainability of the models.

3.2 Discussion

RQ1: Which ethical principles for developing AI-based applications are most often addressed by students?

The master's degree study programme of the students whose dissertations we analyse has only one course dedicated to Ethics and Academic Integrity in Data Science. The course teaches general legal and ethical issues of computer science, IP, software licensing (incl. open source), risks and liabilities, privacy, internet and cyberspace, computer security (hacking). There is also some content on ethical AI, such as data access, use, and collection and ethical aspects of research. These aspects of ethical AI from this course seem to be well addressed by the students in their work.

In terms of fairness and bias, most students address the issues related to unrepresentative data due to class unbalance, synthetic data, the presence of a predominant class or an under-represented class in the dataset. This is specifically acknowledged in 5 of the 7 case studies and is the most predominant, if not the only aspect related to the principle of fairness and bias that is popular. Data annotation issues are also observed, as they may introduce a significant level of subjective interpretation in the data labelling process if not handled properly.

From transparency and explainability, the commonly identified concern is overfitting (in 2 of the 7 papers). In terms of safety and security, only CS1 identifies several cases of possible misuse that can lead to dangerous situations for the user and others around them.

As mentioned previously, one of the works (CS1) has identified a higher number of concerns than others. A possible explanation could be the domain of the paper: automotive, a popular subject of interest in industry, that has already attracted many ethical discussions. Given the advanced technology in this area and the high interest from both industry and research, there are many norms and regulations already in place, generated by ethical implications of self-driving car. The effect is that it increases awareness among the users, developers and researchers alike. This is also perhaps because the safety implications are more direct and more severe than for other topics. Even so, a long-term approach and analysis of ethical implications is necessary in all areas where AI technology is to be frequently used.

In conclusion, most of the students are very familiar with dataset concerns related to dataset containing unbalanced classes. This is probably because they study this aspect specifically in the master's study programme. Also, they are able to identify more ethical concerns in the domains which are highly researched and popular in industry.

RQ2: Which ethical principles for developing AI-based applications are generally neglected by students?

From answering RQ1, we already notice that most ethical concerns are neglected by students in their dissertations. In the area of fairness and bias, there is no concern as to how representative is data to the population (concerning age, ethnicity, gender, etc.), what is

Table 1: Fairness and bias principle related concerns

Concerns	Addressed
CS1	
Speed bumps are less common than other road signs, often datasets lack images with them	YES
Object detection in most models works better with square objects, while speed bumps have uncommon aspect ratio.	YES
Data collection include only Black-Yellow Plastic Speed Bumps in one single metropolitan area, using a single camera setup. The data seems very region specific.	No
One single annotator of data can lead to biased annotation.	YES
The negative class contains images with speed bumps of other shapes/colours and images without speed bumps with background content (unclear what type of content and diversity).	No
Formulations such as ”superior dataset” are misleading without a reference.	No
CS2	
Model testing is based on an existing annotated dataset split on train/validate/test with no description or analysis performed to ensure relevant evaluation and avoid biases.	No
The classes might be imbalanced or unrepresentative for the context.	No
Model classifies all input as fillings because it is a predominant class.	YES
Dental images might not be representative for the entire population (gender, age etc.)	No
CS3	
Utilised pre-trained tool may contain biases, so knowledge on training data is relevant.	No
News content might be unfairly interpreted by sentiment analysis and clustering/aggregation.	No
Statements like ”impressive efficacy of AI-generated summaries” are not supported.	No
CS4	
Dataset is based on MyAnimeList website, with possible user biases (age, region, bots).	No
Classes of anime genres are not balanced.	YES
Some content is never recommended, leading to less popular artists never being promoted.	No
CS5	
Training data is from 1 Jan 2022 to 31 Dec 2023 (520 trading days) and validation data is first quarter of 2024. The period should be representative for trading, not a temporal fluctuation.	No
As a Regulated Market for stock trading, this tool should be equally available to all.	No
CS6	
Data distribution on multiple devices can introduce biases, if unbalanced.	No
Dataset with images but no patient demographics (age, gender).	No
One class presents fewer samples than the others.	YES
Images generated by other miscroscopic devices might not be used in the model.	No
CS7	
Model trained on synthetic data might not be representative.	YES
Access to the app by introducing personal data can lead to fake data.	No
Marketing user profiles might be created to promote specific destinations.	No
Popular destinations might become the only recommendations.	No

the dataset used for the pre-trained models integrated and their biases, or if the data distribution ensures sound evaluation.

The resulting societal implications are drastic: some populations will not get a correct dental diagnosis, summarised news content might add more bias and confusion to the user and influence political developments of a region, several young anime artists will never get a chance to get promoted to the public, the stock market can get even more destabilised by speculating AI models, the use of a different microscope for blood cell imaging might get the wrong diagnosis, and several tourist destinations might get overcrowded while others leave people living on tourism without an income. Fairness is hard to achieve even with the best interests at hand, but if not considered at all it may play havoc on peoples lives, their income, safety, or well-being.

The safety and security risks, as a consequence of system misuse, are also neglected. This is not only related to what an AI model fails to do, but also how that can be exploited by malevolent parties, or simply if the system is used in other ways than intended, and what harm it can do. For example, a false positive

Table 2: Safety and security principle related concerns

Concerns	Addressed
CS1	
Lack of speed bump standardisation migh lead to system not working in a new region.	YES
Speed bump detection could be used to softnes suspention rather than lower speed.	YES
A false positive can make the car unnecessarily brake when no speed bump is present, causing discomfort to the driver and confusing other road users.	YES
A false negative can lead to dangerous situations if driver is overreliant on the system.	YES
At what distance are speed bumps detected (is there enough time to lower speed)?	No
Most false positives are observed on drainage ditches, followed by crosswalks, but also pole shadows or unpaved roads.	YES
Significant false negatives and low recall suggest speed bump is not detected in time, leading to dangerous situations.	No
CS2	
Misdiagnose might exacerbate health issues.	No
System unable to diagnose less frequent issues, as it's trained only on 4 classes.	No
Diagnose does not replace consultation and specialised treatment.	No
CS3	
May increase anxiety, agression or form unbalanced opinions on events.	No
May introduce false elements in summarizing thus spreading misinformation.	No
CS4	
Might promote aggressive or vulgar content.	No
Might promote inappropriate age-group content.	No
CS5	
Economic decisions are still unpredictable, even with a high model accuracy.	No
User might have financial losses.	No
Artificially create market destabilisation due to uniform predictions and decisions.	
CS6	
Can a doctor make safe decisions based on the result of the system? In what cases?	No
A misdiagnosys can be dangerous on the life of a pacient.	No
System requires data security for image upload, diagnoses information.	No
CS7	
System might recommend places with political conflict thus putting user in danger.	No
System might recommend places with sudden bad weather or natural calamities.	No

diagnosis for a leukemia patient can be equivalent to a sentence to death, while a false diagnose of dental cavity is unlikely to cause a person's death, even if it may damage a tooth. In this context, identifying the severity of the risk of a 0.1% error is very important.

The most neglected ethical principle is accountability and liability, with only one concern addressed of the 13 identified. In this regard, the tendency is to leave the whole responsibility to the user of the AI model. While in many cases the user has an important responsibility of the use of a software system, totally neglecting the developer's responsibility is a huge issue at the moment for the whole software engineering world. If software systems and AI models are supposed to be of help and support to humans, it is crucial that they are reliable tools. Thus, developing systems that we can trust is a responsibility of the developer to ensure the data used, training process, and evaluation are properly performed according to rigorous quality standards. Our AI models can generate very high accuracy, but they might be no good if we can not rely on them, or at least guaranteed a level of quality that makes them a reliable support in certain scenarios. Otherwise, the risk is that our AI models are being used as just toys, and not as a trustworthy decision support technology, always with high caution, and as a result they could become more a hindrance than a human support.

Responsibility in ethical AI development usually derives from transparency and explainability. Although some transparency is addressed in the selected case studies, all students completely neglect the explainability principle in their work. All of the projects involve black box models, meaning we have no knowledge whatsoever on how the model makes its decisions. In consequence, the model might go

Table 3: Accountability and liability principle related concerns

Concerns	Addressed
CS1	
Software system has quality standards to be used in automotive independently, and there are people accountable to fulfil them, including updates, model maintenance.	No
Responsible party for using the break or not is the driver.	YES
Responsible party for a misclassification of a speed bump is named (software system quality).	No
CS2	
Software system has quality standards assigned to specific people. The person responsible for a misdiagnosed dental problem is specified.	No
CS3	
Software system has quality standards assigned to specific people. The person responsible for app malfunction (ex. bringing in fake news or discriminating content) is clear.	No
CS4	
The user is responsible for the content they chose to watch, no matter what is the recommendation.	No
Responsibility for the promotion of biased or aggressive or racist content and its repercussions is assigned to a party.	No
CS5	
The user is responsible for their own trading activity and potential financial losses.	No
Companies may be affected by the software predictions and be sabotaged. The responsibility in this case is clear and software quality ensures no misconduct.	No
CS6	
The diagnoses method is informing the doctor, meaning there should be people responsible when the system is at fault (medical device).	No
Software system has quality standards and people accountable to fulfil them, including updates, model maintenance.	No
CS7	
User is responsible for choosing a place to travel, as many conditions are volatile.	No
The system is clear on its limitations of usage.	No

wrong at any time, and we would not know why. It may be because we had an overfitting model, or we have under-represented populations, or the model was not trained well. Not knowing the reasoning of an AI models means we can not rely on its 'guesses'. As many explainability tools have been developed lately, we should encourage their use among computer science students of master's degree level, future software developers, and software engineers designing the AI tools of the future.

RQ3: What solutions for the implementation of ethical AI principles from the literature could be applied in these case studies?

While many of the ethical AI concerns identified deserve a thorough discussion and consideration in a multidisciplinary AI, ethical and law context (Konidena et al., 2024), and many more are yet to be identified in each case study, there are several solutions presented in the literature that can already be applied in these scenarios.

In Figure 7 we present a collection of some representative tools that can be used for ethical AI development. Some are inspired from literature (Cumming et al., 2024), but also online resources such as the Responsible AI Knowledge-base, which is a Git repository of resources developed for this purpose (Alexandra, 2023). This repository has put together a list of Frameworks and Tools for evaluating, explaining and visualising AI models, construction of mimicking models, and research papers. All these have applications for deep neural networks, natural language processing, recommender systems, reinforcement learning, and are accompanied by several specific case studies and examples used in practice. However, some assessments of such tools demonstrate that they are not sufficient to identify subtle and detailed ethical AI concerns (Bubinger and Dinneen, 2024), and we cannot rely on libraries alone. Moreover, ongoing research can help inform developers about which tools to use best for their specific contexts (Jeyakumar et al., 2020).

One of the most common concerns in the development of AI tools is that students (and researchers)

Table 4: Transparency and explainability principle related concerns

Concerns	Addressed
CS1	
Model is black box.	No
Model is overfitting.	No
The user understands the benefits and limitations of the system, how it works, and how to make decisions based on it.	No
Human agency is defined at design level.	No
CS2	
Trains a CNN model - black box.	No
Model is overfitting.	YES
Human agency is specified from the design and clear to the user of the app.	No
Results for each class are presented.	No
CS3	
The accuracy of tools used is not mentioned (and it often falls under 50%).	No
Uses black box models.	No
It is clear to the user how the system selects balanced and comprehensive set of news.	No
CS4	
Black box model requires explanation.	No
User is clear on what they need to do to get an accurate recommendation.	No
CS5	
The model is a combination of network analysis/modelling and decision trees, with some decision making explained.	YES
Overfitting is suspected based on results.	No
It is clear which of the input features (clustering coefficient, degree / closeness / betweenness / eigenvector centrality) are important and how they are used in the prediction model.	No
CS6	
Black box model has no explanation on decision process.	No
Possible overfitting results.	YES
User understands what type of images can be used.	No
CS7	
Black box model.	No
User is informed on how to properly use the app.	No

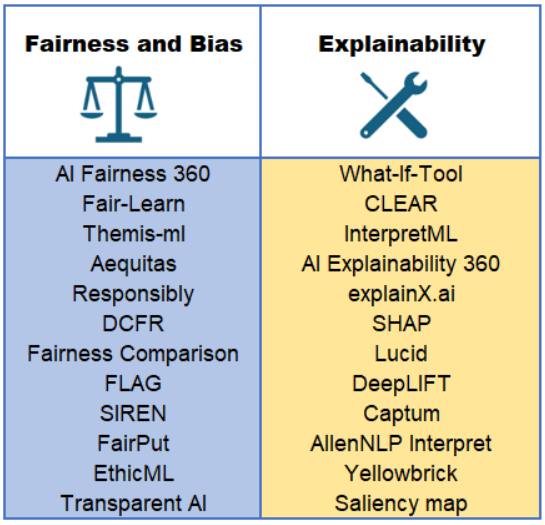


Figure 7: Ethical AI tools.

often first rush into building nice AI models able to solve complex issues, and only second they think about how they fit into society from the industrial point of view. Any design of a software system begins with the intended user and the use cases, and this step would ideally proceed to the implementation. One solution would be to introduce more of the AI Ethics principles for software quality study in the curricula, integrated with AI subjects. In fact, software quality should be addressed as a whole, starting with design principles and requirements of software solutions, whether they use AI or not. Supporting practices for ethical software development have already been proposed for industry: using multi-disciplinary teams, defining the purpose of the AI system, and analyzing the impact of the AI system (Balasubramaniam et al., 2020)

3.3 Study limitations

The number of case studies analysed in this paper is only seven, and might not be sufficiently representative for the general pool of computer science students and AI developers. Researcher bias could also be present in the interpretations and analysis of the ethical aspects of student dissertations. The purpose of this study was not to perform an exhaustive analysis of ethical AI knowledge, considering that this is not yet part of the curricula studies at the master's degree programme, but an assessment used as a starting point. We consider that our findings reflect important aspects of ethical AI literacy and help us gain insight into tomorrow's software developers that design the AI apps we will be soon using and how to intervene for improvement.

4 CONCLUSIONS

AI applications are a significant part of our everyday lives, with important potential to automate and support human activities. However, the frequent use of recent AI tools developed has also revealed many ethical issues hidden in the data used for training or the power of the tools to be used for both good and bad. The software systems have their own autonomy, replacing human action, however, accountability should fall clearly to the responsible for developing, not just using a tool.

This paper analysed 7 master's degree dissertations of students from the specialisation Data Science for Industry and Society from year 2024 in order to assess the level of ethical AI principles addressed by the students. Four different criteria were chosen for the assesment, based on literature review: fairness and bias, safety and security, accountability and liability, and transparency and explainability.

The analysis reveal that the most addressed ethical AI concerns are those related to unbalanced dataset. Explainability is not addressed at all, considering that most works presenting black-box models. The most neglected ethical AI principles are those related to accountability and liability, in which it is expected that the user takes the whole responsibility. Only one of the 7 papers assessed addresses the safety and security concerns of the system developed. Several existing tools for fairness,bias and explainability identified in literature and online resources are recommended both as a support for identifying ethical concerns as well as for mitigation. Another recommendation is the introduction of these principles in the curricula of computer science master's degree programme, closely linked to the subjects related to AI.

In summary, we note that these results emphasise the importance of ethical AI in society and the necessity to integrate ethical AI principles in the various phases of the design and development process of AI models. It is of utmost importance to build a sense of responsibility in developers creating the AI based technology of our future and understand the high impact this has on society in general. The future work proposed involves the development and integration of ethical AI tools to better assess the AI models develop and address the ethical principles described. Also, as a next step, an in depth analysis of the case studies with their general implications in society would present interest and complete this work with more perspectives.

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This section will be completed for the camera ready version. Identifiable information was removed for double-blind review.

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